Software Requirements Specification – Addendum

Implemented Features and Changes since the Initial SRS (SRS-TMS-1.0)

Traffic Management System (Simulation) · Team E16 – Section E · PES University, Bengaluru

Document ID	SRS-TMS-1.1 (Addendum to SRS-TMS-1.0)
Version / Date	1.1 / 12 November 2026
Deliverable	Mini-Project Part 2 – final documentation
Builds on	SRS-TMS-1.0, 29 September 2026

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1 Introduction

1.1 Purpose

This addendum records what the team built against the initial SRS (SRS-TMS-1.0): which planned requirements were implemented, which new features were added that the SRS did not describe, where the implementation interprets or differs from the SRS, and what remains open. It does not replace SRS-TMS-1.0. Requirement IDs (FR, NFR, SEC), use cases (UC) and definitions keep their meaning from that document.

1.2 Sources

The source code and automated tests in the project repository, the GitHub project board (backlog, pull requests), the Test Plan and the Architecture & Design Specification.

2 Summary

- All 12 use cases (UC-01 to UC-12) are realised in the console application (§7).
- Of the 26 requirements of SRS-TMS-1.0 (10 FR, 8 NFR, 8 SEC), **24 are implemented** – three of them with a documented change or interpretation (FR-03, FR-04, FR-06) – and **2 are partially implemented** (NFR-04 and SEC-06, see §6).

- **14 new features** were added that SRS-TMS-1.0 did not describe (§4).
- **9 changes and interpretations** are recorded (§5) and **5 open items** remain (§6).
- Verification: 203 automated tests (unit, integration and scripted whole-program tests), continuous integration with coverage and complexity gates (§8).

3 Status of the Requirements Planned in SRS-TMS-1.0

3.1 Functional requirements

ID	Planned in SRS-TMS-1.0	Status	As built
FR-01	Seeded FIFO queues per lane, 1–60 vehicles/min, one departure per 2 s on green	Implemented	A vehicle arrives in a lane with probability rate/60 each simulated second, drawn from a seeded generator that gives the same sequence on every platform; one departure per 2 s per green lane.
FR-02	Green 5–120 s, yellow 3 s, applies from next cycle	Implemented	Menu 6. The new value shows on the dashboard at once and applies from the next cycle; each change is audited (SET_GREEN).
FR-03	Signal, queue length and congestion per lane, refreshed every simulated second	Implemented, changed	Dashboard per lane (menu 5 and during a run). Refreshed every simulated second at 1× speed; at 10× and 60× one frame per real second (D-01).
FR-04	Emergency priority within 5 s, held until departure, arrival order	Implemented, interpreted	3 s yellow to the open lane, then emergency green; held until the vehicle leaves; simultaneous emergencies served in arrival order. Emergency vehicles are entered by the operator (menu 7, D-02).

ID	Planned in SRS-TMS-1.0	Status	As built
FR-05	Seed, congestion thresholds, session timeout, speed	Implemented	Menu 10 (Administrator). Saved in config.ini; every change is audited (PARAM_CHANGE).
FR-06	Report with throughput per minute, average wait, maximum queue, % High; CSV/TXT within 3 s	Implemented, changed	Menu 8. Throughput is reported as total vehicles and per simulated hour (D-03); an hourly report is written in well under 3 s.
FR-07	Create, deactivate, delete accounts; unique username of 4–20 characters	Implemented	Menu 11: create, list, deactivate, delete (with confirmation).
FR-08	Add, edit, delete intersections: max 10, 1–8 lanes, arrival rate per lane, signal cycle	Implemented	Menu 9. Saved in config.ini. The signal cycle is the green time of the intersection (D-05).
FR-09	Adaptive green extension up to 20 s in 5 s steps, total at most 140 s	Implemented	Adaptive mode is a switch in System parameters (Administrator, off by default). It extends green only while the queue is High and every opposing queue is below Moderate.
FR-10	Commuter view without login, read-only	Implemented	Main menu 2: congestion status, list of completed reports and viewing a report; no function modifies data.

3.2 Non-functional requirements

ID	Planned in SRS-TMS-1.0	Status	As built / evidence
NFR-01	95% of commands within 1 s at 10 intersections × 8 lanes	Implemented	Measured: 95th percentile 0.036 ms, maximum 0.059 ms for 200 commands at the maximum configuration.

ID	Planned in SRS-TMS-1.0	Status	As built / evidence
NFR-02	At least 1,000 ticks/s in fast mode	Implemented	Measured: 3,600 ticks in about 24 ms (about 150,000 ticks/s) at the maximum configuration.
NFR-03	24 simulated hours without crash, hang or more than 10% memory growth	Implemented	Automated reliability tests at the maximum configuration, including a 24-simulated-hour run (issue #18). The unit tests also ran clean under AddressSanitizer and UBSan at integration.
NFR-04	4 of 5 first-time testers start a simulation within 5 min; menus at most 3 levels	Partially implemented	On-screen help on every menu (h), menus at most three levels deep (automated check) and a scripted first-time-user test (TC-17). The trial with five testers is an open item (O-2).
NFR-05	Header/source pairs; complexity at most 10; coverage at least 70%	Implemented	No function above complexity 10 (average 3.0 over 260 functions at integration); line coverage 96.5% at integration; both limits are enforced in CI.
NFR-06	Invalid input rejected with field name and range	Implemented	Every prompt names the field and range (for example: Choice must be between 0 and 11); empty input or input over 256 characters is refused and state is unchanged.
NFR-07	Zero warnings with g++ -Wall -Wextra on Ubuntu 22.04 and Windows 11	Implemented, stricter	Built with -Wall -Wextra -Wpedantic -Werror: CI on Ubuntu 22.04, team machines on Windows 11 with MinGW-w64 (D-09).

ID	Planned in SRS-TMS-1.0	Status	As built / evidence
NFR-08	Same seed gives identical reports	Implemented	Byte-identical reports for the same seed at the maximum configuration; a different seed gives a different report (automated tests).

3.3 Security requirements

ID	Planned in SRS-TMS-1.0	Status	As built
SEC-01	Login for every Operator/Administrator function; Commuter read-only	Implemented	Every menu action passes the Command Dispatcher; without login only the Commuter view is available.
SEC-02	Password of 8+ characters with letter and digit; salted hash; no plaintext	Implemented, changed	Policy enforced (8–256 characters). PBKDF2-HMAC-SHA256, 10,000 iterations, 16-byte random salt (D-06). Passwords are never stored or logged.
SEC-03	60 s lock after 3 consecutive failed logins	Implemented	Attempts 1–3 give E-AUTH-001; the 4th attempt, even with the right password, gives E-AUTH-002; login works again after 60 s (see D-07).
SEC-04	Session ends after inactivity (default 10 min)	Implemented	Default 10 minutes, configurable 1–60 (menu 10). An expired session gives E-AUTH-003 and returns to the login screen.
SEC-05	Role-based access control on every command; denied command changes nothing	Implemented	One central permission check; a denied command shows E-AUTH-004 and is audited as DENIED.

ID	Planned in SRS-TMS-1.0	Status	As built
SEC-06	Input limits, no unbounded C functions, hardening flags, cppcheck without high findings	Partially implemented	Input limited to 256 characters and range-checked; std::string is used instead of unbounded C functions. The flags -fstack-protector-strong and -D_FORTIFY_SOURCE=2 and a cppcheck run are not yet part of the build (O-1).
SEC-07	Audit record for logins, lockouts, user and parameter changes, emergency overrides, exports; no edit or delete	Implemented	audit.log is append-only with no function to edit or delete; events are listed in N-10.
SEC-08	users.dat and audit.log owner-only; report names reject separators and ..	Implemented (Linux)	Mode 0600 on Linux; report names reject path separators and .. (see O-4 for Windows).

4 New Features Added Beyond SRS-TMS-1.0

ID	Feature	Description	Related
N-01	First-start administrator setup	If no account exists, the program asks for the first administrator account. No default password is built in.	UC-01, UC-07, SEC-02
N-02	Example intersection	On first start one example intersection (4 lanes) is created so a simulation can be started at once.	FR-08
N-03	Run, pause, resume, stop with live dashboard	Run the simulation for 1–3,600 simulated seconds with the dashboard refreshed once per real second; pause, resume and stop at any time.	UC-04, FR-03

ID	Feature	Description	Related
N-04	Emergency vehicle injection	The operator places an emergency vehicle in a chosen lane of a chosen intersection; the signal gives it priority.	UC-08, FR-04
N-05	Adaptive timing switch	Administrator switch in System parameters; off by default; saved in config.ini and audited; Operators cannot change it.	UC-09, FR-09
N-06	On-screen help	Pressing h at any menu prompt shows help for that screen.	NFR-04
N-07	Account deactivation and guarded deletion	Deactivated accounts cannot log in and are marked in the list; deletion asks for confirmation; an administrator cannot remove their own account; older user files still load.	FR-07
N-08	Commuter report viewer	The Commuter can list and open completed reports by name; unsafe names are refused and files that are not reports are hidden.	FR-10
N-09	Error-code catalogue	Every refusal shows its code with the message, for example [E-AUTH-004], [E-VAL-001], [E-IO-002].	NFR-06

ID	Feature	Description	Related
N-10	Extended audit trail	LOGIN_OK, LOGIN_FAIL, ACCOUNT_LOCKED, LOGIN_BLOCKED, SESSION_EXPIRED, LOGOUT, INITIAL_ADMIN_CREATED, USER_CREATE, DENIED:(command), SET_GREEN, PARAM_CHANGE, CONFIG_CORRUPT, SIM_START, SIM_PAUSE, SIM_RESUME, SIM_STOP, EMERGENCY_INJECTED, EMERGENCY_CLEARED, REPORT_EXPORT; user deactivation, deletion and the adaptive switch are audited as well.	SEC-07
N-11	Safe handling of damaged files	A damaged config.ini gives E-CFG-001 and default settings; the file is left untouched so it can be repaired. A damaged user file stops start-up with a message instead of running unprotected.	SO-3
N-12	Report safeguards	A report is never overwritten; the extension .csv or .txt is added automatically; names are limited to 64 characters and refuse path separators, .., drive and wildcard characters; refused exports are audited too.	FR-06, SEC-08
N-13	Authentication hardening	Unknown usernames cost the same as wrong passwords and keep no state (no memory growth, no account guessing); constant-time hash comparison; audit fields cannot forge extra log lines.	SO-3, SO-1

ID	Feature	Description	Related
N-14	Engineering quality gates	GitHub Actions builds with warnings as errors, runs all tests and enforces coverage of at least 70% and complexity of at most 10 on every push and pull request; scripted whole-program tests drive the real program; performance, reliability and repeatability tests.	NFR-01 to NFR-08

5 Changes and Interpretations

ID	SRS reference	Planned	As built	Reason
D-01	FR-03	Dashboard refreshed at least once per simulated second	At 1× every simulated second; at 10× and 60× one frame per real second (every 10th or 60th simulated second).	Printing 60 screens per second is unreadable in a console.
D-02	FR-04	Detect an emergency vehicle within 1 tick of entering a lane	Emergency vehicles enter through the operator command (menu 7) and are detected in the same tick.	The SRS does not define where emergency vehicles come from.
D-03	FR-06, §1.3	Throughput in vehicles per simulated minute	Report gives total vehicles and vehicles per simulated hour (per minute = per hour divided by 60).	Hourly figure is clearer for one-hour runs.
D-04	FR-06	Percentage of time at High congestion	A simulated second counts as High if at least one lane is High.	Definition needed for several lanes and intersections.

ID	SRS reference	Planned	As built	Reason
D-05	FR-02, FR-08	Signal cycle	One lane open at a time in lane order; the cycle is the green time per intersection (5–120 s).	The SRS does not define signal phases.
D-06	SEC-02	SHA-256 with at least 10,000 iterations, or Argon2id	PBKDF2-HMAC-SHA256 with 10,000 iterations and a 16-byte random salt.	Standard construction of an iterated SHA-256 hash.
D-07	SEC-03	60 s lock after 3 failed logins	Implemented as specified; the failure count and lock time are kept in memory, so restarting the program clears a lock.	Simpler user file; stronger variant is an optional improvement.
D-08	Test Plan data	Test users op1 and op2	Test users oper1 and oper2.	FR-07 requires usernames of 4–20 characters; op1 has 3.
D-09	NFR-07	Warnings checked with -Wall -Wextra	Also -Wpedantic and -Werror.	Stricter quality bar; compiler warnings fail the build.

6 Open Items and Limitations

ID	Item	Related
O-1	Add -fstack-protector-strong and -D_FORTIFY_SOURCE=2 to the build and run cppcheck in CI, aiming at zero high-severity findings.	SEC-06
O-2	Record the usability trial with five first-time testers (4 of 5 must start a simulation within 5 minutes).	NFR-04

ID	Item	Related
O-3	Optionally store failed-login count and lock time in users.dat so a lock survives a restart.	SEC-03, D-07
O-4	On Windows the owner-only restriction of users.dat and audit.log relies on the permissions of the user profile folder.	SEC-08
O-5	Passwords are visible on screen while typed (standard console input).	SO-1

7 Use-Case Status

Use case	Where in the program
UC-01 Log In	Main menu 1
UC-02 Monitor Traffic Signals	Operations menu 5 and the live dashboard during a run
UC-03 Control Signal Timing	Operations menu 6
UC-04 Simulate Traffic Flow	Operations menus 1–4
UC-05 Manage Intersections	Operations menu 9
UC-06 Configure System Parameters	Operations menu 10 (Administrator)
UC-07 Manage User Accounts	Operations menu 11 (Administrator)
UC-08 Handle Emergency Vehicle Priority	Operations menu 7
UC-09 Apply Adaptive Signal Timing	Adaptive switch in menu 10
UC-10 / UC-11 Generate and Export Reports	Operations menu 8
UC-12 View Congestion Status	Main menu 2 (Commuter, no login)

8 Verification Summary

- **Automated tests:** 203 tests run with the project build (make test) and in GitHub Actions: unit tests per module, integration tests of signals with the engine, and scripted whole-program tests that type menu choices into the real program.
- **Test Plan cases covered by scripted tests:** TC-01 (accounts and password policy), TC-02 (lockout and unlock), TC-03 (session timeout), TC-04 (Operator refused Administrator functions), TC-13 (Commuter cannot reach control functions), TC-17 (first-time user starts a simulation and reads lane 2 with help only).
- **Quality gates:** line coverage of the program code 96.5% (1,480 of 1,533 lines at integration, limit 70%); no function above cyclomatic complexity 10 (limit 10); build with -Wall -Wextra -Wpedantic -Werror.
- **Memory and undefined behaviour:** the unit tests ran without findings under AddressSanitizer and UBSan at integration.
- **Performance:** 95th percentile command time 0.036 ms (limit 1 s); about 150,000 ticks/s in fast mode (limit 1,000).
- **Repeatability:** two runs with the same seed give byte-identical reports.